

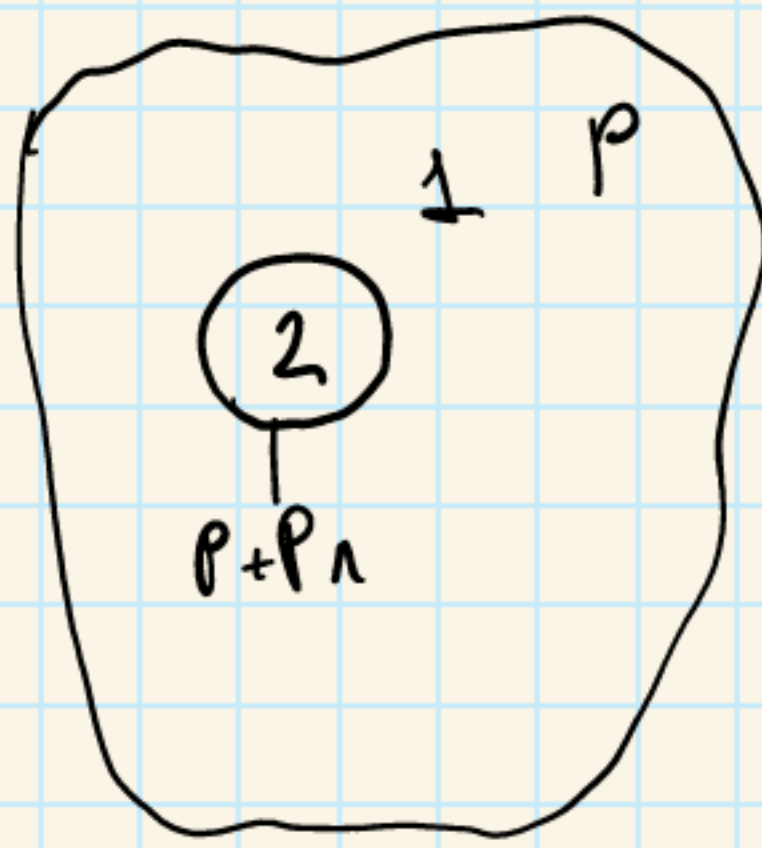
$$\left(\frac{\partial P}{\partial V}\right)_S = \frac{\partial(P, S)}{\partial(V, S)} \cdot \frac{\partial(V, T)}{\partial(P, T)} = \frac{\partial(P, S)}{\partial(P, T)} \cdot \frac{\partial(V, T)}{\partial(V, S)} =$$

$$= \left(\frac{\partial S}{\partial T}\right)_P \cdot \left(\frac{\partial T}{\partial S}\right)_V = \frac{C_P}{C_V}$$

$$\left(\frac{\partial P}{\partial V}\right)_S = \gamma \left(\frac{\partial P}{\partial V}\right)_T$$

Зав-ть равновесного давления от К
(кривизна)

$$K = \frac{1}{R_1} + \frac{1}{R_2}$$



$$P_1(P) = P_2(P + P_L)$$

$$P_0 + \frac{\partial P_1}{\partial P} \Delta P = P_0 + \frac{\partial P_2}{\partial P} (\Delta P + P_L)$$

$$dP = -\sigma d\gamma + \gamma dK$$

$$\Delta P = \sigma_2 (\Delta P + K \sigma)$$

$$\Delta P = K \sigma \frac{\sigma_2}{\sigma_1 - \sigma_2}$$

1) Пузырёк пара в жидкости

$$\sigma_2 \gg \sigma_1$$

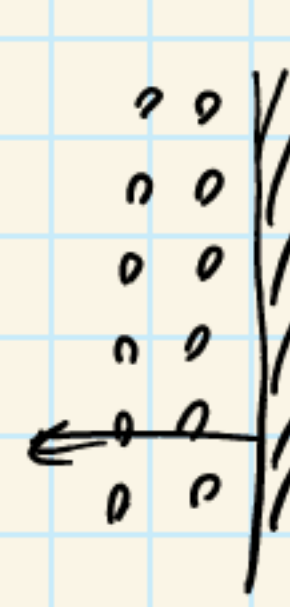
$$\Delta P = -K \sigma = -\frac{2\sigma}{R}$$

2) Капля в паре

$$\sigma_2 < \sigma_1$$

$$\Delta P = \frac{2\sigma}{R} \cdot \frac{\sigma_2}{\sigma_1}$$

Модель идеального газа.

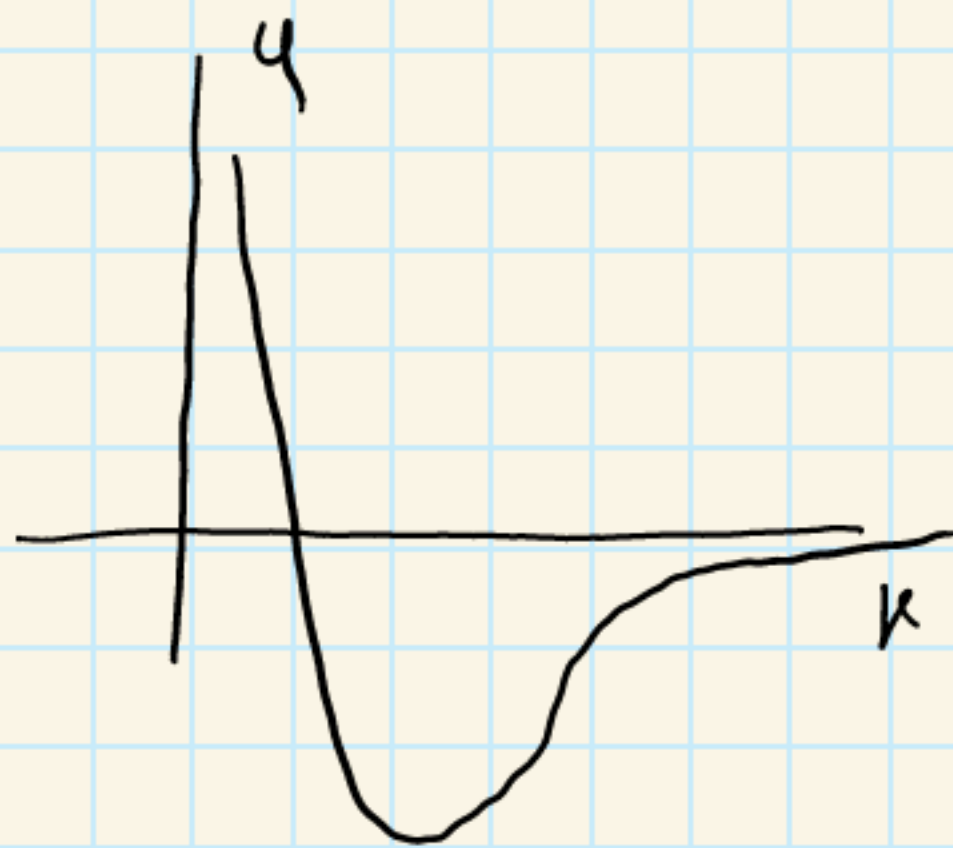


$$PV = RT$$

$$P(V - b) = RT$$

$$\left(P + \frac{\gamma^2 a}{V^2}\right)(V - \gamma b) = \gamma RT$$

Потенциал Леннард-Джонса



$$U = \epsilon \left[\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 \right]$$

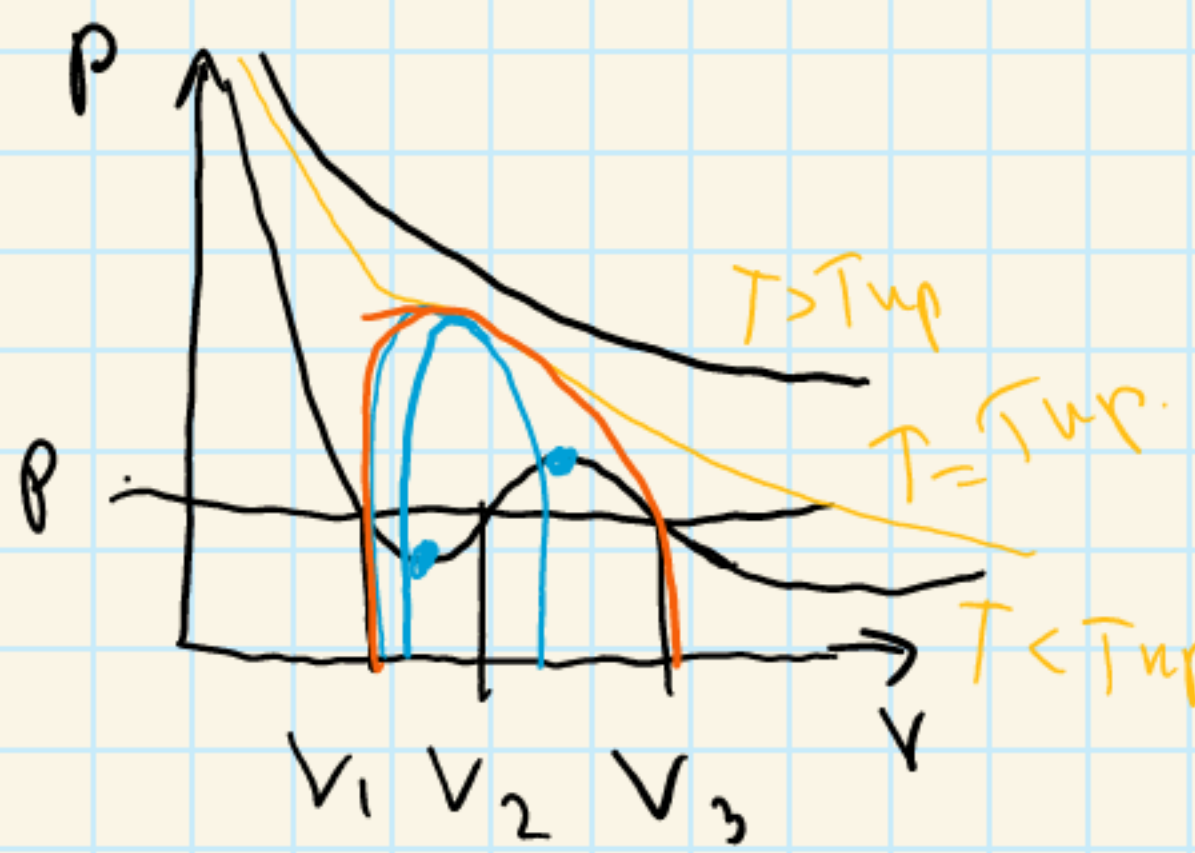
γ_r - Коэффициент - Окекса

$$P = RT \left(\frac{\gamma}{V} + \beta_2 \frac{\gamma^2}{V^2} + \beta_3 \frac{\gamma^3}{V^3} + \dots \right)$$

β_i - вириальные коэф.

Изотермы газа Ван-дер-Ваальса

$$(PV^2 + a)(V - b) = V^2 RT - \gamma_r \text{ - е 3й степени отн. } V$$



$$\left(\frac{\partial P}{\partial V}\right)_T = 0$$

кривая Спинодаль

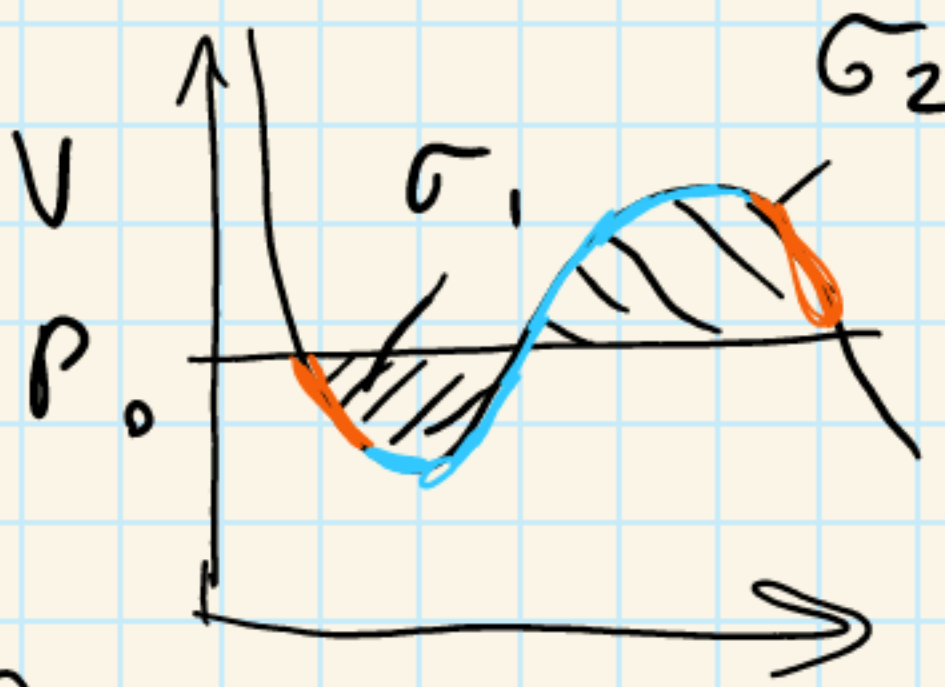
бинодаль, $P_1 = P_2$

$$P_1 = P_2$$

$$\int_{V_1}^{V_2} dP = 0$$

$$\int_{V_1}^{V_2} [-\sigma d\gamma + \gamma dK] = \int_{V_1}^{V_2} \gamma dP =$$

$$= \int_{V_1}^{V_2} (P(V) - P_0 dV), \quad V dP = d(PV) - P dV$$



$$\Rightarrow \sigma(PV) - \int P dV = P_0(V_2 - V_1) - \int P(V) dV = 0$$

$\sigma_1 = \sigma_2$ - правило Максвелла.

Критические параметры

$$PV^3 - (Pb + RT)V^2 + aV - ab = 0$$

$$P(V - V_{кр})^3 = 0$$

$$P_{кр} V^3 - 3P_{кр} V_{кр} V^2 + 3P_{кр} V_{кр}^2 V - P_{кр} V_{кр}^3 = 0$$

$$3P_{кр} V_{кр} = P_{кр} b + RT_{кр}$$

$$V_{кр} = 3b, \quad P_{кр} = \frac{a}{27b^2}, \quad T_{кр} = \frac{8a}{27bR}$$

$$\frac{P_{кр} V_{кр}}{RT_{кр}} = \frac{3}{8} \approx 0,375$$

ка самот Ziele

$$0,26 \div 0,3$$

$$\varphi = \frac{V}{V_{кр}}, \quad \pi = \frac{P}{P_{кр}}, \quad \tau = \frac{T}{T_{кр}}$$

$$\left(\pi + \frac{3}{\varphi^2}\right)\left(\varphi - \frac{1}{3}\right) \approx \frac{8}{3} \tau - \text{кривые Ван-Дер-Ваальса.}$$